

From: Leonard, Micky (DEQ)
Sent: 6/15/2018 2:42:48 PM
To: "Brady, Donal (DEQ)" <BradyD6@michigan.gov>
Cc: "McElmurry, Ashley (DEQ)" <McElmurryA@michigan.gov>
Subject: Cadillac Final Report
Attachments: 2018 Cadillac WWTP CSI Final Report.xlsx

Don,

Please find attached the Final Report for the 2018 CSI of Cadillac WWTP. Please take some time to review it and let me know if there are any questions or errors and I will correct those items. Once it looks good please upload it to MiWaters and send a copy on to the facility.

Thanks so much and don't hesitate to contact me with any questions or concerns.

Cheers,

Micky Leonard
248-763-1635
Water Resources Division
Southeast Michigan District Office



ATTACHMENT NAME:

2018 Cadillac WWTP CSI Final Report.xlsx

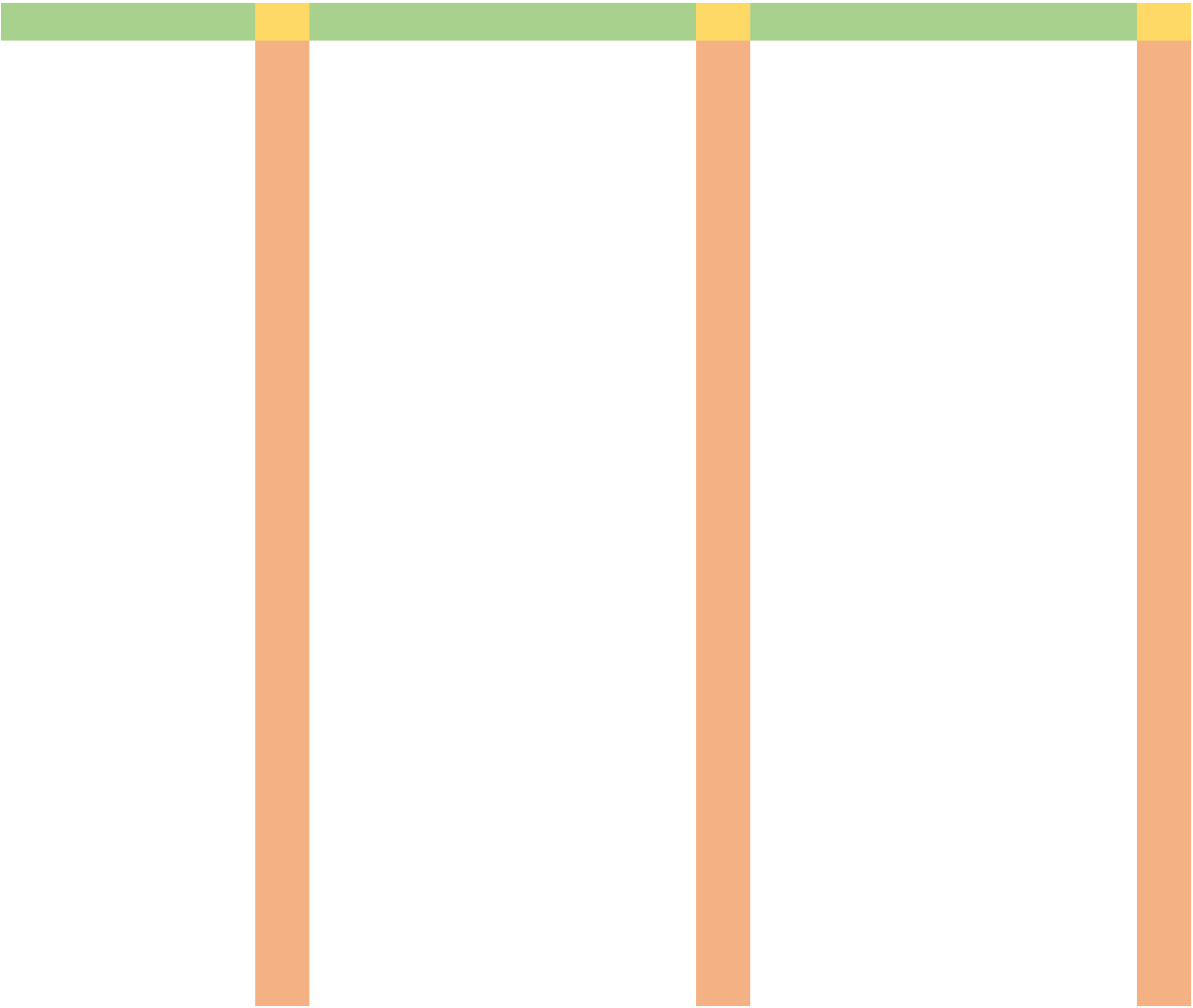
ATTACHMENT TYPE:

Microsoft Office 12+, Office Open XML (OOXML/OpenXML): spreadsheet (XLSX)

Water Composite 1		Det Lev	Water Grab 1 -1
Facility Name	Cadillac WWTP		Cadillac WWTP
Monitoring Point	001A		001A
Survey Period Start Date/time	5/9/2018 10:20		
Survey Period End Date/time	5/10/2018 9:20		
Flow Rate (MGD)	1.975		
Grab Matrix			Wastewater
Grab Date Collected			5/9/2018
Grab Time Collected			10:20
Parameters	Results		Results
Temperature (F)			61
pH (S.U.)			6.99
TRC			
Dissolved Oxygen			8.06
Conductivity (umho/cm)			
Fecal Coliform			---
Oil and Grease			
	Results	Units	Results
Suspended Solids	7	mg/L	<4 ND
Dissolved Solids		mg/L	<20 ND
CBOD-5		mg/L	<3 ND
BOD-5		mg/L	
COD		mg/L	
TOC		mg/L	
NO2		mg/L	<0.01 ND
NO3 + NO2		mg/L	<0.01 ND
NH3	0.12	mg/L	<0.01 ND
Kjeldahl Nitrogen		mg/L	<0.1 ND
Total Phosphorus	0.15	mg/L	<0.01 ND
Ortho Phosphate		mg/L	<0.01 ND
Total Cyanide		µg/L	<5 ND
Amenable Cyanide		µg/L	
Available Cyanide		mg/L	<0.002 ND
Alkalinity (as CaCO3)	130	mg/L	<20 ND
Hardness (as CaCO3)		mg/L	<1 ND
Calcium		mg/L	<1 ND
Sodium		mg/L	<1 ND
Potassium		mg/L	<0.1 ND
Magnesium		mg/L	<1 ND
Chloride		mg/L	<1 ND
Sulfate		mg/L	<2 ND
Sulfide		mg/L	

	Location	Parameter	Unit	Result	Limit
x		Total Arsenic (As)	ND	µg/L	<1 ND
x		Total Barium (Ba)	23	µg/L	<5 ND
x		Total Cadmium (Cd)	ND	µg/L	<0.2 ND
x		Total Chromium (Cr)	1.2	µg/L	<1 ND
x		Total Copper (Cu)	6.4	µg/L	<1 ND
x		Total Lead (Pb)	ND	µg/L	<1 ND
x		Total Mercury (Hg)	ND	µg/L	<0.2 ND
x		Total Selenium (Se)	ND	µg/L	<1 ND
x		Total Silver (Ag)	ND	µg/L	<0.2 ND
x		Total Zinc (Zn)	20	µg/L	<5 ND
		Total Aluminum (Al)		µg/L	<5 ND
		Total Antimony (Sb)		µg/L	<1 ND
		Total Beryllium (Be)		µg/L	<1 ND
		Total Boron (B)		µg/L	<20 ND
		Total Cobalt (Co)		µg/L	<5 ND
		Hexavalent Chromium (Cr+6)		µg/L	<5 ND
		Total Iron (Fe)		µg/L	<20 ND
		Total Lithium (Li)		µg/L	<10 ND
x		Total Nickel (Ni)	4.5	µg/L	<2 ND
		Total Manganese (Mn)		µg/L	<5 ND
x		Total Molybdenum (Mo)	12	µg/L	<5 ND
		Total Strontium (Sr)		µg/L	<5 ND
		Total Thallium (Tl)		µg/L	<2 ND
		Total Titanium (Ti)		µg/L	<5 ND
		Total Vanadium (V)		µg/L	<2 ND
		Phenolics		µg/L	

Det Lev	Water Grab 1 - 2	Det Lev	Water Grab 1 - 3	Det Lev
	Cadillac WWTP 001A		Cadillac WWTP 001A	
	Wastewater		Wastewater	
	5/9/2018 15:35		5/10/2018 9:20	
Units	Results	Units	Results	Units
	59		62	
	6.8		8.17	
	8.44		8.33	
(cts/100 ml) <10 ND	---	<10 ND	ND	<10 ND
Units	Results	Units	Results	Units



Upstream	Det Lev
Cadillac WWTP Upstream	

Surface Water
5/9/2018
11:00

Results	Units
62	
7.06	
9.05	

Results	Units



***MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY
WATER RESOURCES DIVISION
POINT SOURCE MONITORING***

Report of a
Municipal Wastewater Survey
Conducted at
Cadillac WWTP
NPDES No. MI0020257
Wexford County
Cadillac, MI 49601
May 9-10, 2018

SURVEY SUMMARY

Wastewater monitoring was performed during one, twenty-four hour survey period starting May 9, 2018 at 10:20 AM.

The results of the physical, chemical and bacteriological analyses are presented in Tables 1 & 2.

The results of this survey are compared to the results reported by the facility in Tables 1 & 2.

A composite sample from monitoring point 001A was split with the facility to compare laboratory analyses. The results are shown in Table 3.

The results of this survey are compared to the results of the previous survey performed in October of 2014 in Table 4.

Samples were preserved according to Table 5. Letter codes for laboratory results are defined in Table 5.

SURVEY PROCEDURE

The flow and samples were obtained as follows:

<u>Sample Location</u>	<u>Flow Measurement</u>	<u>Sampling Methods</u>
001A - Composite samples collected prior to disinfection from facility compositor	Facility Analysis Report	Grab composite samples
001A - Grab samples collected post disinfection from outfall to Clam River	---	Individual grab samples
Upstream samples collected from approximately 20 ft. upstream of outfall to Clam River	---	Individual grab samples

A grab composite consists of a series of individual grabs composited into one sample.

An individual grab is a single instantaneous sample.

Samples were analyzed by the MDEQ Laboratory located in Lansing, Michigan, and Merit Laboratories, Inc. in East Lansing, MI.

	MDEQ Analyses		Cadillac WWTP Analyses	
Monitoring Point	001A		001A	
Survey Period Start Date/Time	5/9/18 10:20		5/9/18 7:00	
Survey Period End Date/Time	5/10/18 9:20		5/10/18 7:00	
Flow Rate (MGD)	2.14		2.14	
	Results		Results	
	<u>mg/L</u>	<u>lbs/day</u>	<u>mg/L</u>	<u>lbs/day</u>
Suspended Solids	7	124.933	4	71.390
CBOD-5	<3 ND	NA	9.31	166.161
NH3	0.12	2.142	0.556	9.923
Total Phosphorus	0.15	2.677	0.111	1.981
Available Cyanide	<0.002 ND	NA	---	---
Alkalinity (as CaCO3)	130	2320.188	---	---
	<u>µg/L</u>	<u>lbs/day</u>	<u>µg/L</u>	<u>lbs/day</u>
Total Arsenic (As)	<1 ND	NA	---	---
Total Barium (Ba)	23	0.410	---	---
Total Cadmium (Cd)	<0.2 ND	NA	---	---
Total Chromium (Cr)	1.2	0.021	---	---
Total Copper (Cu)	6.4	0.114	---	---
Total Lead (Pb)	<1 ND	NA	---	---
Total Mercury (Hg)	<0.2 ND	NA	---	---
Total Selenium (Se)	<1 ND	NA	---	---
Total Silver (Ag)	<0.2 ND	NA	---	---
Total Zinc (Zn)	20	0.357	---	---
Total Nickel (Ni)	4.5	0.080	---	---
Total Molybdenum (Mo)	12	0.214	---	---

Monitoring Point Matrix Date Collected Time Collected	MDEQ Analyses				Cadillac WWTP Analyses	
	Grab 1-1	Grab 1-2	Grab 1-3	Upstream	Day 1	Day 2
	001A	001A	001A	Upstream	001A	001A
	Wastewater	Wastewater	Wastewater	Surface Water	Wastewater	Wastewater
	5/9/2018	5/9/2018	5/10/2018	5/9/2018	5/9/2018	5/10/2018
	10:20	15:35	9:20	11:00	7:00	7:00
			<u>Results</u>		<u>Results</u>	
Temperature (F)	61	59	62	62	---	---
pH (S.U.)	6.99	6.8	8.17	7.06	7.08	6.94
Dissolved Oxygen (mg/L)	8.06	8.44	8.33	9.05	8.02	7.65
Fecal Coliform (cts/100 ml)	---	---	<10 ND	---	1	<1 ND
Available Cyanide (mg/L)	---	---	<0.002 ND	---	---	---

Composite Split

Monitoring Point

001A

Date Collected

5/10/2018

Time Collected

9:20

	MDEQ Results	Facility Results
	<u>mg/L</u>	<u>mg/L</u>
Suspended Solids	6	4
NH3	0.44	0.556
Total Phosphorus	0.16	0.111
Alkalinity (as CaCO3)	130	---
	<u>µg/L</u>	<u>µg/L</u>
Total Arsenic (As)	<1 ND	---
Total Barium (Ba)	24	---
Total Cadmium (Cd)	<0.2 ND	---
Total Chromium (Cr)	1.1	---
Total Copper (Cu)	7.6	---
Total Lead (Pb)	<1 ND	---
Total Mercury (Hg)	<0.2 ND	---
Total Selenium (Se)	<1 ND	---
Total Silver (Ag)	<0.2 ND	---
Total Zinc (Zn)	20	---
Total Nickel (Ni)	4.7	---
Total Molybdenum (Mo)	13	---

Monitoring Point	001A	
Matrix	Wastewater	
Survey Period Start Date/Time	10/14/14 10:15	5/9/18 10:20
	<u>Results</u>	
	<u>mg/L</u>	<u>mg/L</u>
Suspended Solids	7	7
CBOD-5	4.8	<3 ND
NH3	0.31	0.12
Total Phosphorus	0.15	0.15
Available Cyanide	---	<0.002 ND
Alkalinity (as CaCO3)	130	130
	<u>µg/L</u>	<u>µg/L</u>
Total Arsenic (As)	---	<1 ND
Total Barium (Ba)	---	23
Total Cadmium (Cd)	---	<0.2 ND
Total Chromium (Cr)	---	1.2
Total Copper (Cu)	7.3	6.4
Total Lead (Pb)	---	<1 ND
Total Mercury (Hg)	---	<0.2 ND
Total Selenium (Se)	---	<1 ND
Total Silver (Ag)	---	<0.2 ND
Total Zinc (Zn)	---	20
Total Nickel (Ni)	---	4.5
Total Molybdenum (Mo)	---	12

Parameter	Preservative
COD/TOC/Phenol/Nutrients (Chlorine Absent)	5 drops H ₂ SO ₄ /250 ml (to pH <2)
Available Cyanide	Preserved with Sodium Hydroxide to a minimum pH of 12.
pH & Dissolved Oxygen (D.O.)	Meter reading
Total Metals	2 ml 1:1 HNO ₃ /250 ml (to pH <2).
Microbiology	2 drops 10% sodium thiosulfate/125 ml to dechlorinate sample

Samples preserved as required, cooled to 6 degrees Centigrade with chain of custody maintained

Lab Letter Codes

ND - Indicates compound analyzed for but not detected

Survey by: Micky Leonard, Environmental Quality Analyst

Contact with Management: Doug Langworthy, Superintendant Cadillac WWTP

Laboratory Analyses by: Michigan Department of Environmental Quality Laboratory,
Lansing and Michigan Merit Laboratories Inc., East Lansing,
Michigan

Report by: Micky Leonard, Environmental Quality Analyst
Field Operations Section
Water Resources Division
Southeast Michigan District Office

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